Intelligent Document Processing Platform

Software Requirement Specification (SRS)

GITHUB LINK: https://github.com/juweriya1/ai-document-processing

BY

Team Lead: ***Fatima Naeem 26933***

Member(s): ***Maheen Muhammad Rizwan 27118***

Juweriya Bint Nasir 26787

PROJECTS COMMITTEE (PC)

SUPERVISOR: ***Dr. Rizwan Ahmed Khan***

SUBMITTED TO

PROJECTS Manager – FYP

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IBA SMCS

School of Mathematics and Computer Science

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Software Requirement Specification (SRS)

1. Problem Statement

Enterprises generate large volumes of structured and unstructured documents in multiple formats, including PDFs, Word files, and scanned images. Existing solutions like template-based OCR and LLM-based approaches fail to address multi-format input, privacy concerns, hallucination, and deterministic extraction. Manual processing is time-consuming and prone to errors, and enterprise workflows require reliable, accurate, and actionable insights.

Our project proposes an **Intelligent Document Processing Platform** that ingests documents, extracts data using OCR and NLP/ML techniques, validates against schemas, allows human-in-the-loop feedback, stores information securely in a database, and generates predictive analytics and dashboards for decision-making. This ensures high accuracy, privacy, and reliability in enterprise document management.

2. System Requirement

2.1. List of Functional Requirements

FR #	Functional Requirement	Description
FR1	Multi-format Document Ingestion	Support PDFs, Word files, scanned images, handwritten text, multilingual.
FR2	Pre-processing	Deskew, clean, resize images for OCR accuracy
FR3	OCR and Layout Analysis	Extract text, bounding boxes, segment text/tables/figures
FR4	Entity Extraction	Identify invoice numbers, dates, vendors, line items, clauses, total .
FR5	Schema Validation	Validate extracted data against predefined rules.
FR6	Human-in-the-Loop Feedback	Users can review and correct extracted data
FR7	Database Integration	Store and retrieve validated data securely (SQL/NoSQL).
FR8	Analytics & Dashboards	Generate spend analysis,

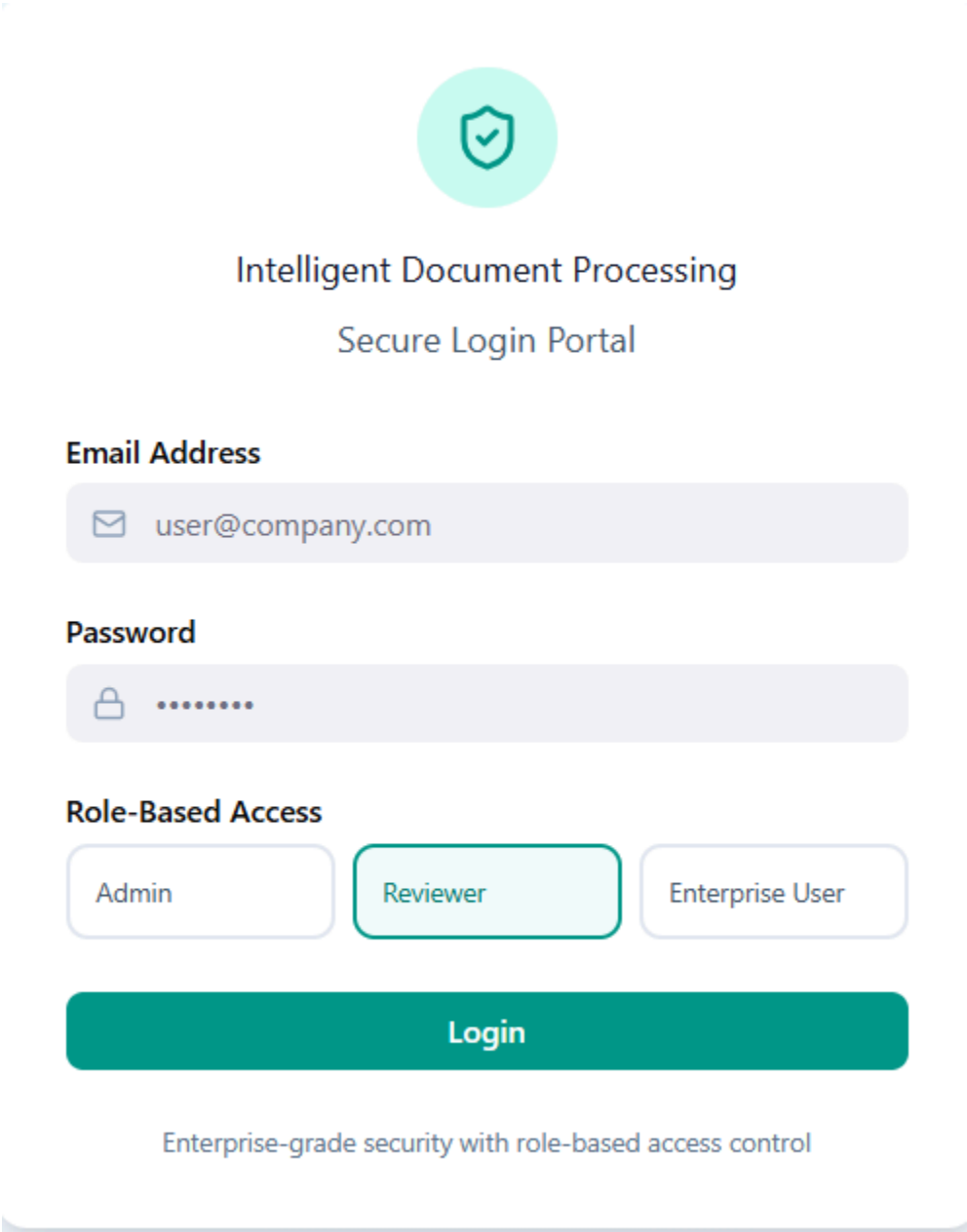
		supplier performance, compliance dashboards
FR9	Predictive Modeling	Forecast supplier delays, anomalies, and compliance risks.
FR10	Security & Privacy	Role-based access, secure processing environment.

2.2. List of Non-functional Requirements

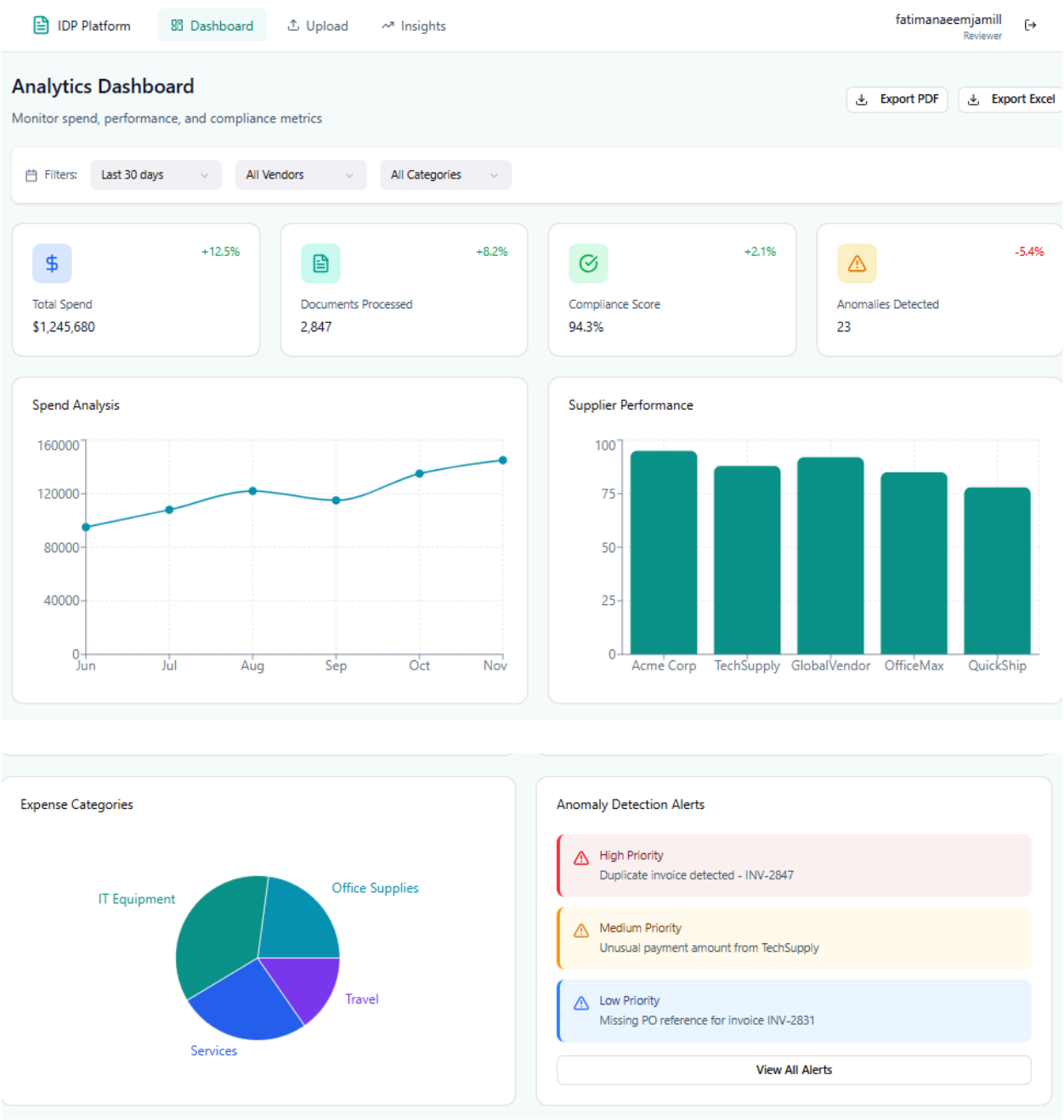
NFR #	Non-functional Requirement	Description
NFR1	Accuracy	≥95% extraction accuracy for structured data.
NFR2	Reliability	Deterministic output, minimal errors.
NFR3	Scalability	Process hundreds of documents per day.
NFR4	Performance	Optimized OCR + extraction pipeline speed.
NFR5	Maintainability	Modular design for future improvements.
NFR6	Security & Compliance	Enterprise privacy standards for sensitive documents.
NFR7	Usability	User-friendly interface for correction and dashboard access.

2.3. User Interface Requirements

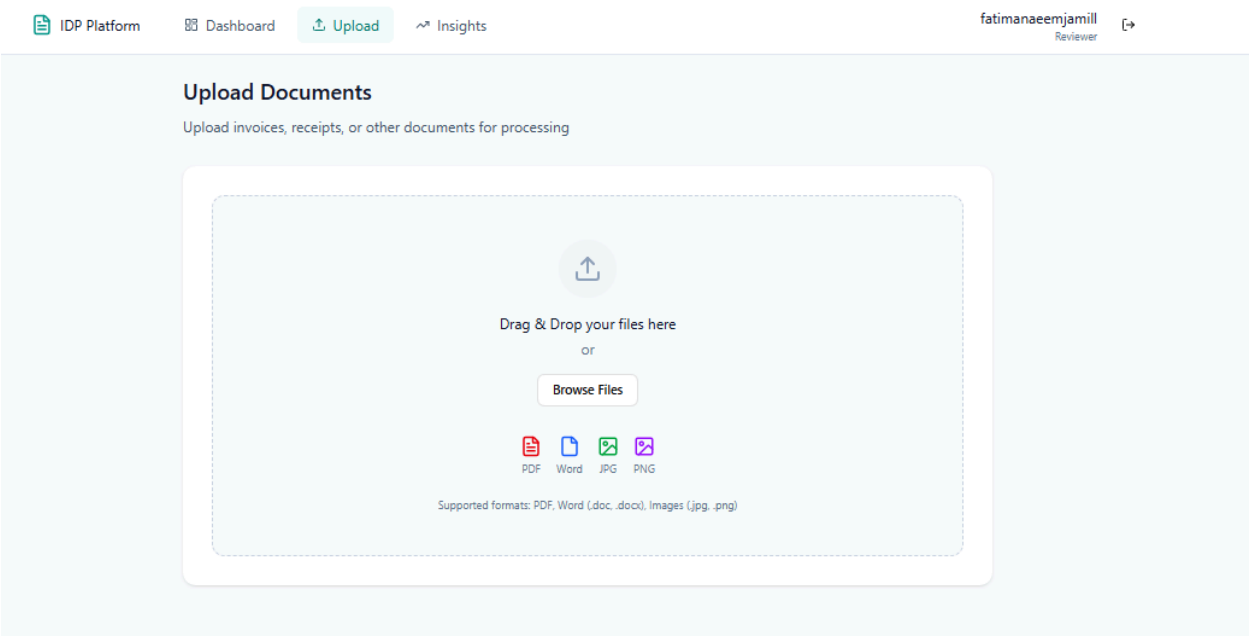
LOGIN PAGE



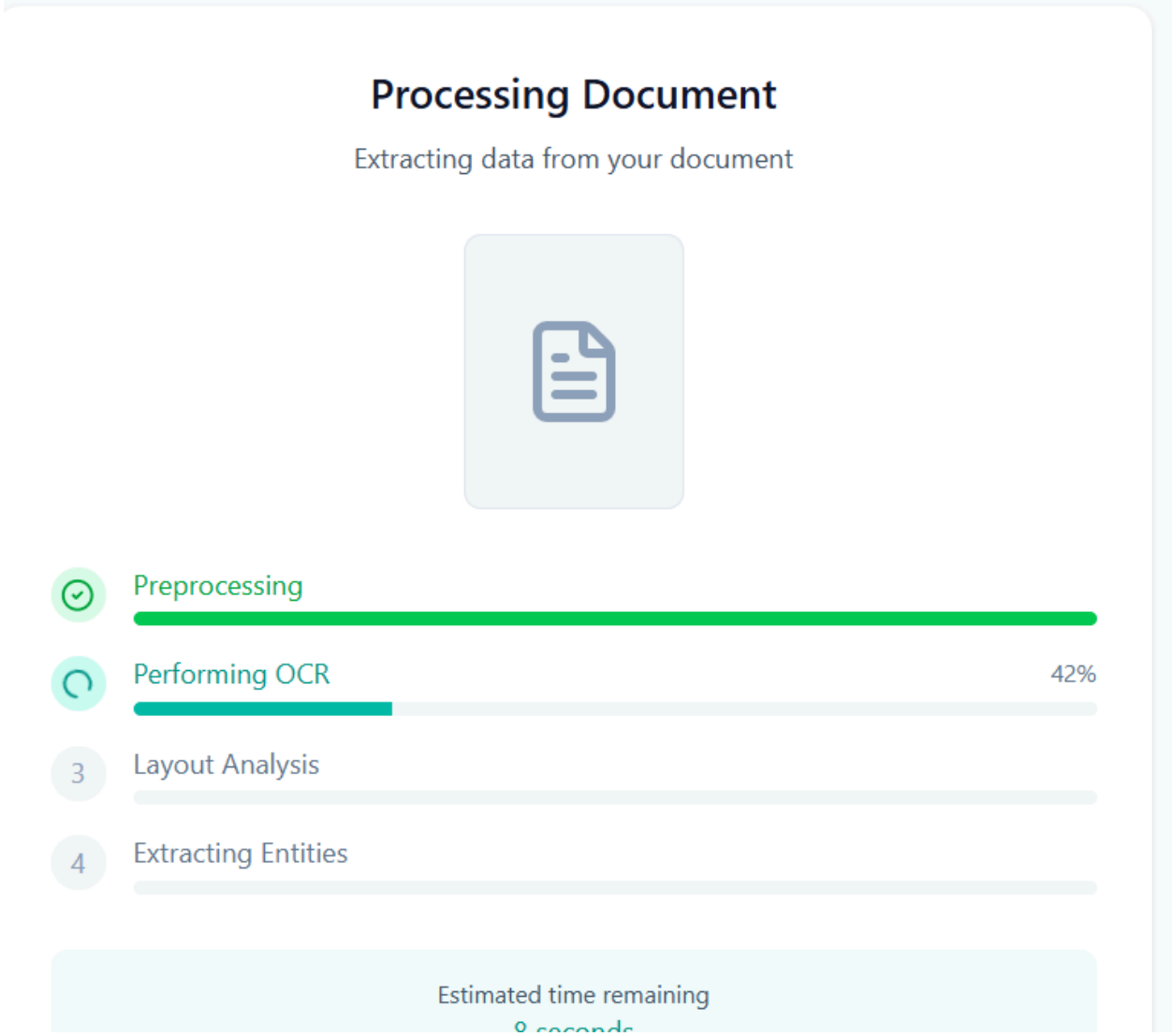
DASHBOARD PAGE



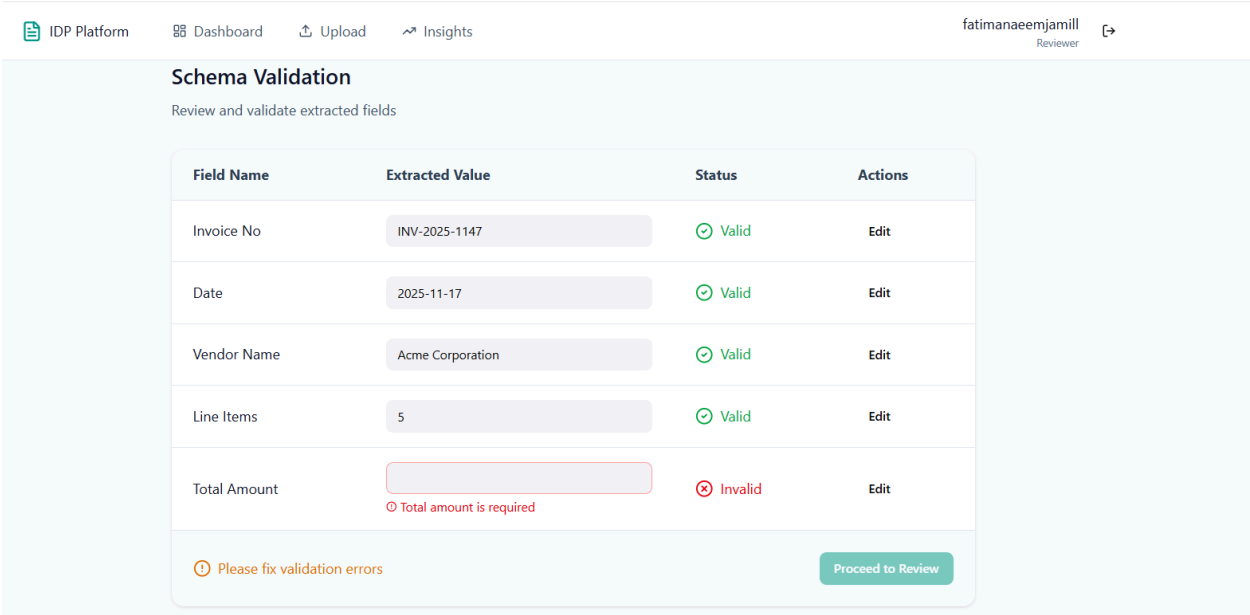
UPLOAD DOCUMENTS PAGE



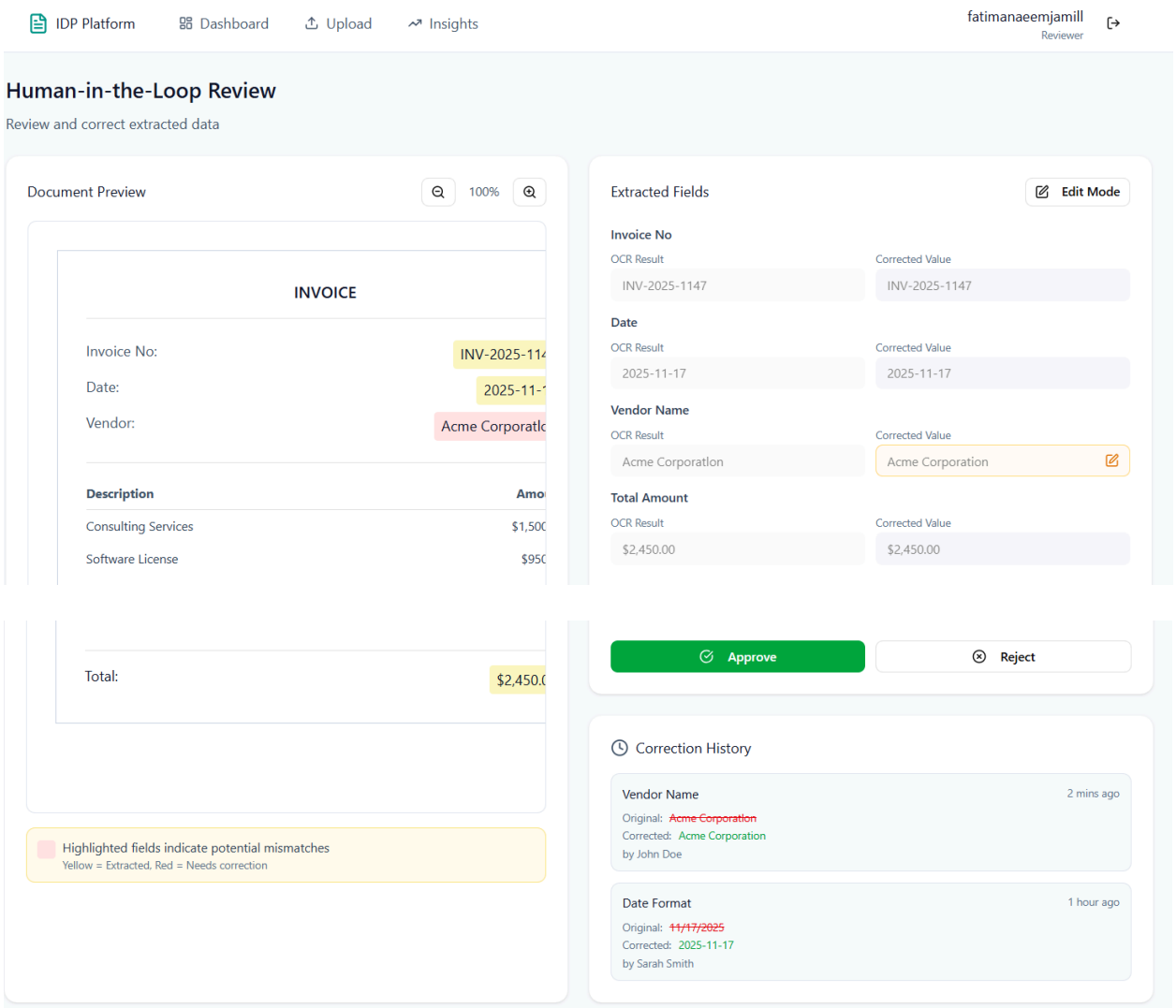
PREPROCESSING



SCHEMA VALIDATION PAGE



HUMAN IN LOOP PAGE



INSIGHTS PAGE

IDP Platform | Dashboard | Upload | Insights

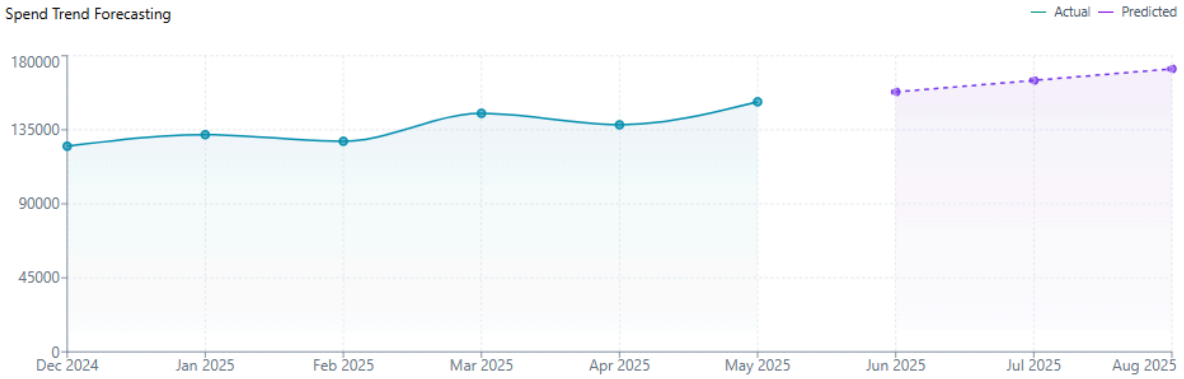
fatimanaeemjamill
Reviewer

Predictive Insights

AI-powered forecasting and risk analysis

Download Prediction Report

Spend Trend Forecasting



Supplier Delay Risk Score

TechSupply Inc. — High Risk — 78 Risk Score

Risk Factors:
- Late deliveries (3 times last month)
- Payment delays increasing
- Quality complaints up 15%

GlobalVendor Corp — Medium Risk — 65 Risk Score

Risk Factors:
- Seasonal demand fluctuations
- Minor delivery inconsistencies

Acme Corporation — Low Risk — 25 Risk Score

Risk Factors:
- Consistent on-time delivery
- Strong quality metrics

AI-Generated Insights

Predicted Spend Increase
Based on historical patterns, spending is expected to increase by 12% in Q3 2025
Confidence: 87% Impact: High

Supplier Consolidation Opportunity
Analysis suggests consolidating 3 vendors could reduce costs by $45,000 annually
Confidence: 92% Impact: High

Seasonal Trend Detected
Office supply purchases peak in September, recommend bulk ordering in August
Confidence: 95% Impact: Medium

How are these predictions generated?

Historical Data Analysis
Our AI analyzes 12+ months of transaction history, identifying patterns, seasonality, and trends in your spending behavior.

Machine Learning Models
Advanced regression and time-series models predict future outcomes with confidence intervals based on statistical significance.

Risk Assessment
Supplier performance metrics, delivery history, and quality indicators are combined to calculate comprehensive risk scores.

Note: Predictions are estimates based on historical data and should be used as guidance. Actual outcomes may vary due to market conditions, organizational changes, and other external factors.

- Dashboard for analytics and predictive insights.
- Human-in-the-loop review panel for verifying extracted data.
- Document upload interface supporting multiple formats.
- Secure login and role-based access.
- Reports: PDFs, tables, and charts for compliance, spending, and supplier analysis

3. Functional Requirements Specification

3.1. Stake Holders

Stakeholder	Interest / Role in System
Enterprise Users (Finance, Procurement, Legal Teams)	Upload documents, review extracted data, generate reports, use dashboards.
Human Reviewer / Analyst	Verify extracted information and correct errors (human-in-the-loop).
System Administrator	Manage user accounts, assign roles, ensure system security, maintain databases.
Project Developers / Engineers	Build and maintain OCR, extraction, validation, and analytics modules.
Supervisors / Advisors	Review system design and deliverables, ensure compliance with FYP requirements.

3.2. Actors and Goals

Actor	Goals
Enterprise User	• Upload documents • View extracted data • Approve or review output • Access analytics dashboard
Human Reviewer	• Ensure extracted data is correct

	• Edit wrong fields • Provide feedback to improve system accuracy
ML Extraction Engine (System Actor)	• Perform OCR • Extract entities • Validate data against schema
System Administrator	• Manage users and permissions • Monitor activity logs • Maintain database and backups

3.3. Use Cases

Below are the primary use cases derived from the Functional Requirements (FR1–FR10):

Use Case 1: Upload Document

- **Actor:** Enterprise User
- **Goal:** Upload a PDF, Word file, or scanned image into the system.
- **Summary:** User selects a document, system preprocesses it, and sends it to OCR.

Use Case 2: Run OCR & Extract Entities

- **Actor:** ML Extraction Engine
- **Goal:** Automatically extract text, tables, and key information such as dates, vendor names, invoice numbers, totals, etc.
- **Summary:** After upload, system performs OCR and layout analysis.

Use Case 3: Validate Extracted Data

- **Actor:** System
- **Goal:** Ensure extracted fields match predefined schema (e.g., valid date, numeric amount).
- **Summary:** System checks extracted values before saving.

Use Case 4: Human-in-the-Loop Review

- **Actor:** Human Reviewer
- **Goal:** Review extracted fields, correct errors, and approve results.

- **Summary:** System displays extracted data; reviewer edits fields and submits corrections.

Use Case 5: Store Data in Database

- **Actor:** System
- **Goal:** Save clean and validated data to secure storage.
- **Summary:** After approval, structured data is stored in SQL/NoSQL DB.

Use Case 6: View Dashboard and Analytics

- **Actor:** Enterprise User
- **Goal:** See spend analysis, supplier performance, anomaly detection, etc.
- **Summary:** System displays visual reports and trends.

Use Case 7: Generate Predictions

- **Actor:** ML Engine
- **Goal:** Forecast supplier delays, anomalies, or compliance risks.
- **Summary:** Models analyze stored data and generate predictions.

Use Case 8: Manage Users and Roles

- **Actor:** System Administrator
- **Goal:** Control access, assign roles, and maintain system security.
- **Summary:** Admin creates accounts, sets permissions, and monitors logs.

4. User Interface Specification

This section describes the user interfaces associated with each use case. The system provides a set of interactive screens for document upload, review, validation, analytics, and prediction. Each screen is designed to be simple, intuitive, and aligned with the functional requirements of the system.

4.1 Upload Document

The user is presented with an upload interface where they can submit files for processing. The interface includes:

- A drag-and-drop upload area
- A button to browse and select files from the device
- A file type indicator (PDF, Word, JPG, PNG supported)

- A progress indicator showing upload status
- Buttons for “Upload” and “Cancel”

This UI enables users to ingest documents in multiple formats quickly and conveniently.

4.2 OCR & Extraction Processing

After uploading a document, the user sees a processing screen that displays:

- OCR progress bar
- Status messages (e.g., “Extracting text…”, “Performing layout analysis…”)
- Document preview thumbnail
- Estimated time to complete processing

This gives visibility into the extraction workflow while maintaining user engagement.

4.3 Schema Validation

Once extraction is completed, the system automatically validates the extracted fields. The schema validation interface includes:

- A list of extracted fields (invoice number, dates, amounts, vendor name, etc.)
- Validation indicators (✓ valid, ✗ invalid)
- Error messages for fields that do not match schema rules
- A button to proceed to the human review screen

This UI ensures transparency of validation checks before data is finalized.

4.4 Human-in-the-Loop Review

The human reviewer sees a dedicated review panel consisting of:

- Document preview on the left
- Editable extracted fields on the right
- Buttons to Approve, Edit, or Reject the extraction
- A section showing previous corrections made (for system learning)
- Highlighted mismatches between OCR output and schema expectations

This interface allows users to manually correct errors and ensures high accuracy.

4.5 Save to Database

A confirmation interface is shown once the reviewer approves the extraction:

- A summary of validated fields

- A confirmation message: “Data successfully stored”
- Options to return to dashboard or upload another document

This UI gives users clear feedback that their data is securely saved.

4.6 Dashboard & Analytics

The system dashboard presents visual analytics and summary reports:

- Navigation tabs for Spend Analysis, Vendor Performance, Anomalies, and Trends
- Charts such as bar charts, line graphs, and pie charts
- Summary cards showing totals, averages, and performance metrics
- Filters for date range, vendor, category, etc.
- Export options (PDF, Excel)

This UI provides a comprehensive view of enterprise document insights.

4.7 Prediction Output

The prediction screen displays insights generated by ML models:

- Supplier delay risk score (Low, Medium, High)
- Expected delivery delays or anomalies
- Explanation of factors contributing to prediction
- Option to download prediction reports
- Button to view historical trend comparison

This interface enables users to make data-driven decisions.

4.8 User & Role Management

The admin panel includes:

- A table listing all users with their roles and status
- Controls to add a new user, edit a role, or deactivate an account
- Filters for active/inactive users
- Logs for activity tracking and security monitoring

This UI supports secure administration and access control.

5. Domain Analysis

The domain of this project is **Intelligent Document Processing (IDP)**, which involves automated extraction, validation, storage, and analysis of information from enterprise documents such as invoices, contracts, receipts, and purchase orders. The system integrates

OCR, NLP/ML models, rule-based schema validation, human feedback, and predictive analytics.

The following domain model illustrates the major entities, data flows, and relationships within the system

5.1 Domain Model

Domain Entities

1. **Document**
   Input provided by the user (PDF, Word, scanned image).

2. **OCR Engine**
   Converts visual document content into machine-readable text with bounding boxes.

3. **Layout Analyzer**
   Segments document into logical components (text blocks, tables, sections).

4. **Entity Extractor (ML + Rules)**
   Extracts structured fields (invoice number, dates, amounts, vendor, items, clauses).

5. **Schema Validator**
   Checks extracted data against predefined business rules and formats.

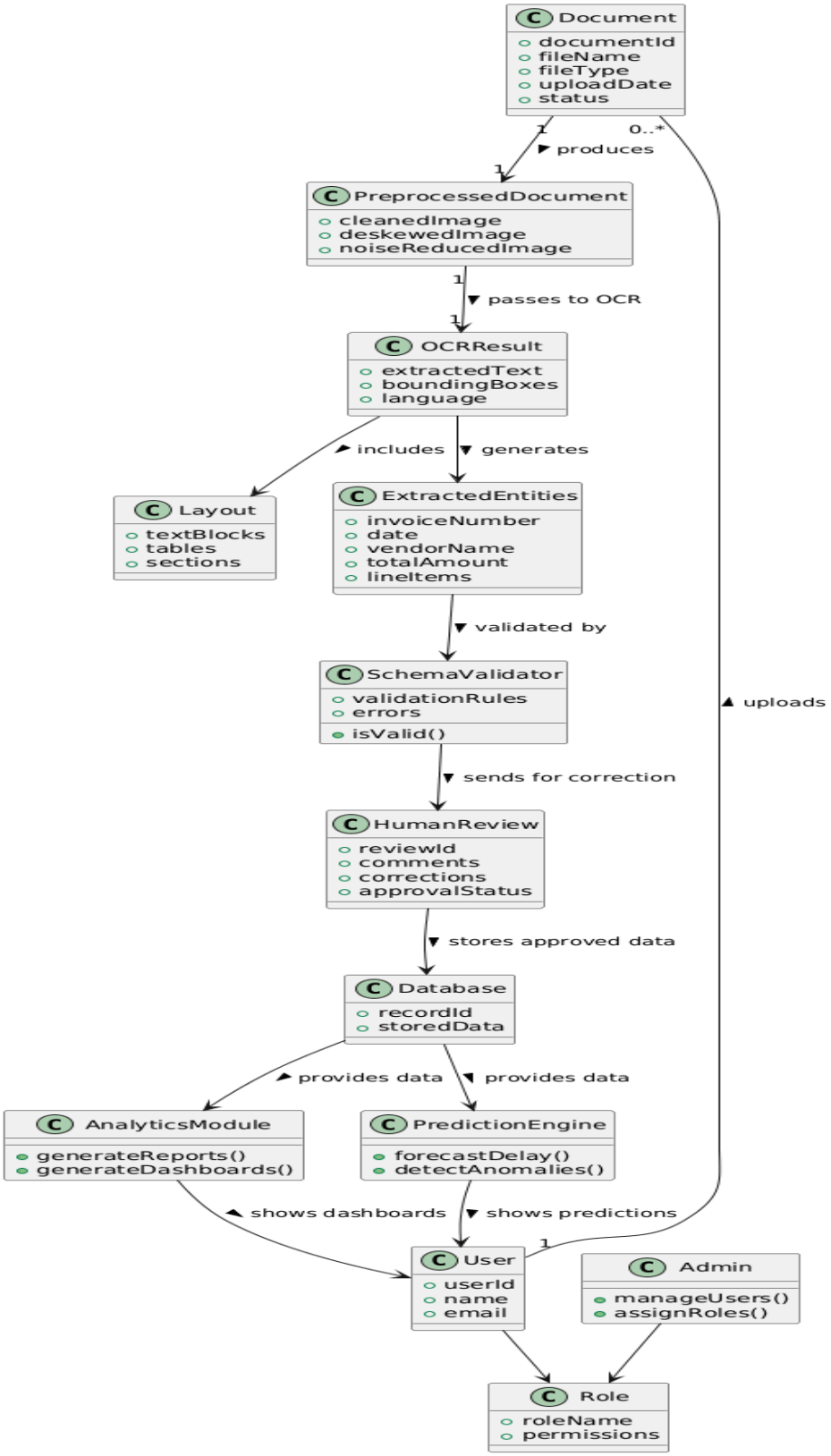
6. **Human Reviewer**
   Reviews extracted data, corrects errors, and approves final output.

7. **Database**
   Stores validated and cleaned information.

8. **Analytics & Reporting Module**
   Generates dashboards, summary reports, and insights.

9. **Prediction Engine (ML)**
   Provides supplier risk predictions, anomaly detection, and trend forecasts.

5.2 Domain Model Diagram



5.3 Mathematical / Computational Models Used

Our project includes multiple analytical and ML components:

1. OCR Model (Vision Transformer or Tesseract OCR)

- Converts document pixel values $I(x, y) \rightarrow$ text + layout.

- Recognizes characters via probability:
  $c\hat{} = arg\, max c\, P(c|I)$

2. Layout Aware LLM for Structured Extraction

Our improved system uses a multimodal LLM to combine text, image regions, and layout embeddings.

Multimodal fusion:

$$h = \mathrm{Fuse}(h_{\text{text}}, h_{\text{image}}, h_{\text{layout}})$$

Token generation:

$$P(x_t \mid x_{<t}, D)$$

Instruction-conditioned extraction:

$$y = f_\theta(D, \text{instruction})$$

Where y contains structured fields such as:

- Vendor
- Invoice Number
- Amount
- Date
- Line-Item Table

3. Named Entity Recognition (NER) Model

Used for extracting:

- Invoice Number
- Date

- Vendor Name
- Amount
- Line Items

NER uses sequence labeling:

$$P(y \mid x) = \prod_{i=1}^{n} P(y_i \mid x, y_{<i})$$

Where:

- x = document text
- y = entity labels (DATE, TOTAL, VENDOR)

4. Rule-Based Schema Validation

Ensures determinism and accuracy:

- Date format check:

$$IsValidDate(x) = (x\, matches\, regex\, YYYY - MM - DD)$$

- Amount validation:

$$Total = \sum(Line\ Items)$$

5. Predictive Analytics Models

To forecast supplier delays, anomalies, etc.

Common models used:

- **Random Forest**
  $y\hat{} = mode(h1(x), h2(x), ..., hk(x))$
- **Time-series forecasting (LSTM/ARIMA)**
  Predicts future trends in expenses or delivery times.

- **Anomaly Detection**

$$\text{Anomaly}(x) = \begin{cases} 1 & \text{if } |x - \mu| > k\sigma \\ 0 & \text{otherwise} \end{cases}$$

6. Human-in-the-Loop Reinforcement

Corrections improve extraction accuracy over time:

$\theta new = \theta old - \eta \nabla \theta (L(predicted, corrected))$

6. Conclusion

This Software Requirement Specification (SRS) defines the complete functional, non-functional, interface, and domain requirements for the Intelligent Document Processing System. By integrating OCR, machine learning, schema validation, human-in-the-loop correction, secure storage, analytics, and predictive modeling, the system aims to deliver a reliable, accurate, and scalable solution for enterprise document automation. The requirements outlined in this document provide a structured foundation for the system's design, development, and implementation, ensuring that the final product meets the needs of stakeholders while maintaining high standards of security, usability, and performance.